

$$\begin{array}{ccc}
\mathbb{C}_{\text{can}}^N & \xrightarrow{\text{Split}_k} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{can}}^n \\
\downarrow \text{C2S}_N & & \downarrow \bigoplus_{r=0}^{k-1} \text{C2S}_n \\
\mathbb{C}_{\text{coeff}}^N & \xrightarrow{S} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{coeff}}^n \\
\downarrow f^{\oplus N} & & \downarrow \bigoplus_{r=0}^{k-1} f^{\oplus n} \\
\mathbb{C}_{\text{coeff}}^N & \xrightarrow{S} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{coeff}}^n \\
\downarrow \text{S2C}_N & & \downarrow \bigoplus_{r=0}^{k-1} \text{S2C}_n \\
\mathbb{C}_{\text{can}}^N & \xleftarrow{\text{Merge}_k} & \bigoplus_{r=0}^{k-1} \mathbb{C}_{\text{can}}^n
\end{array}$$